

THE ABELIAN SCALING BOTTLENECK, AND THE ANGULAR HURWITZ SYSTEM: WHERE NONCOMMUTATIVE ARITHMETIC THERMODYNAMICS CAN AND CANNOT LIVE

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ABSTRACT. The series has twice failed to produce a genuinely noncommutative arithmetic thermodynamics: the free semigroup (Paper V) loses the high-temperature phase to orthogonal range projections, and the Hurwitz quaternions (Papers X, XII) have nested ranges but a scalar dynamics driven by the reduced norm, which factors through the abelianization. We prove that the second failure is universal — the *abelian scaling bottleneck*: for a right LCM semigroup with scalar one-parameter dynamics $\sigma_t(v_a) = N(a)^{it}v_a$, the KMS classification data splits into a scaling measure governed by the P^{ab} -graded weight N and a β -independent tracial datum on the isotropy; every β -driven transition is a transition of the abelianized weight data, and no non-abelian residual symmetry group can be produced on the semigroup side — consistently with Paper VII's finding that its essential-value group is abelian, and verified concretely on the braid monoids B_n^+ , right LCM and maximally noncommutative, whose entire transition is that of the length series while the braiding survives only as β -independent traces. The complementary half is Paper XXVII: noncommutativity lives *transverse* to the scaling, as a cocycle into a compact group, realizing $\text{Prob}(H \backslash G)$. We then name the Goldilocks candidate the two halves point to: the *angular Hurwitz system*, whose radial part Nrd carries the thermodynamics of Paper X ($\beta_c = 2$) and whose discarded angular part $\rho(a) = a/\text{Nrd}(a)^{1/2} \in SU(2)$ is an ordered cocycle in the sense of Paper XXVII. The $\pm a$ symmetry of the representation sets makes the matrix Kakutani series exactly computable, $\Xi_\beta(\rho) = 96 \sum_p (p+1)p^{-\beta}$ up to the projective normalization, with abscissa *exactly* the radial critical exponent 2: the angle turns critical precisely where the radius does. The programme — transport of Paper XXVII's matrix martingale to the Hurwitz tail relation in the Gibbs phase, expected verdict $\text{KMS}_\beta\text{-simplex} \cong \text{Prob}(SU(2))$ for $\beta > 2$, with the Lubotzky–Phillips–Sarnak spectral gap closing the other side — is stated with its single hard step named.

1. THE BOTTLENECK THEOREM

Let P be a countable right LCM monoid with trivial units for simplicity, $N : P \rightarrow \mathbb{R}_{>0}$ a multiplicative weight with finite level sets, $\mathcal{T}(P)$ its Nica–Toeplitz algebra with generating isometries v_a , nested range projections, and the scalar dynamics $\sigma_t(v_a) = N(a)^{it}v_a$.

Lemma 1.1 (The trivial half, stated because it is the point). $\mathbb{R}_{>0}$ is abelian, so N factors through the abelianization P^{ab} : the Gibbs weight of every monomial $v_a v_b^*$ depends only on the classes of a, b in P^{ab} .

Theorem 1.2 (Abelian scaling bottleneck). *Let φ be a σ -KMS $_\beta$ state on $\mathcal{T}(P)$ (or any gauge-invariant quotient). Then:*

- (1) *On the fixed-point algebra of the gauge action, spanned by $v_a v_b^*$ with $N(a) = N(b)$, φ is a trace; in particular on every subalgebra generated by N -preserving symmetries (the “kernel directions”) the state is tracial, with no β -dependence in the defining condition.*

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Paper XXXIV of the series on localized Bost–Connes systems. The framework is that of [1], cited by DOI; Papers V, VII, X, XII, XXVII and XXVIII of the series are referred to by numeral, and every fact used from them is restated in place.

- (2) φ is determined by the pair (its restriction to the diagonal, a scaling measure with cocycle $N^{-\beta}$; its tracial field on the isotropy), in the Neshveyev parametrization [4]; the scaling measure sees only the P^{ab} -graded counting data of N .
- (3) Consequently the β -driven structure of the KMS simplex — existence ranges, uniqueness, symmetry breaking — is that of an abelianized weight system, and the residual symmetry acting on extreme states through the scaling-measure component is a quotient of a group of characters: no scalar dynamics on a right LCM semigroup produces a nonabelian residual symmetry group on that component.

Proof sketch. (1) is the two-line computation used throughout the series (Paper XXVII, its Theorem 2.1): for x in the fixed algebra and analytic y there, KMS gives $\varphi(xy) = \varphi(y\sigma_{i\beta}(x)) = \varphi(yx)$ since σ is trivial on the fixed algebra. (2) is the groupoid form of the KMS condition: the diagonal restriction is quasi-invariant with Radon–Nikodym cocycle $N^{-\beta}$, whose value on any groupoid element depends only on the P^{ab} -class by Lemma 1.1, and the fibre datum over the isotropy is a field of tracial states, weightless because the cocycle is trivial on isotropy of N -preserving type; extremality decomposes accordingly [4]. (3) follows: two extreme states with the same measure component differ by a β -independent tracial datum, and the action of any symmetry on the measure component is through the characters detected by the $N^{-\beta}$ -weighted series of [Main], an abelian dual object. $\square$

Example 1.3 (Braid monoids: maximal noncommutativity, abelian transition). The braid monoid B_n^+ is right LCM (Garside theory [2]) and strictly noncommutative for $n \geq 3$; its abelianization is $\mathbb{Z}$, generated by word length ℓ , and any scalar dynamics is $\sigma_t(v_a) = q^{it\ell(a)}$ for some $q > 1$. By Theorem 1.2 the entire KMS transition structure is that of the growth series $\sum_a q^{-\beta\ell(a)}$ — a radius-of-convergence phenomenon on $\mathbb{Z}$, with at most circle-character symmetry — while the braiding survives only in the β -independent tracial fibre. The Goldilocks checklist (noncommutative, nested, coupled) fails at the third item, and the theorem says it must.

Remark 1.4 (Consistency across the series). Theorem 1.2 retro-explains three findings: Paper VII’s essential-value group E_β came out abelian because it had to; Paper XII’s reduction of the search to the norm-one subgroup located the noncommutativity exactly in the weightless kernel directions, where item (1) makes it tracial; and Paper XXVIII’s no-go then showed those tracial directions are, for arithmetic orders, hyperfinite. The bottleneck is the common cause.

2. THE COMPLEMENTARY HALF, AND WHERE THE GRAIL LIVES

The bottleneck forbids noncommutativity in the *scaling* direction. Paper XXVII proved it thrives *transverse* to it: keeping the scalar dynamics and the principal tail relation, an ordered Frobenius cocycle c into a nonabelian compact group G yields KMS simplices $\text{Prob}(H \setminus G)$, H the Mackey range, with genuinely nonabelian breaking, lift-dependence, and the matrix Kakutani criterion (its Theorem 4.3). The two statements are the two halves of one verdict: *scaling abelian, symmetry arbitrary*. The grail, if it exists, is a single arithmetic semigroup whose canonical structure produces such a cocycle intrinsically.

3. THE ANGULAR HURWITZ SYSTEM

The Hurwitz monoid P of Paper X discards, in its dynamics, exactly half of each generator: $a = \text{Nrd}(a)^{1/2} \cdot \rho(a)$ with

$$\rho(a) := a/\text{Nrd}(a)^{1/2} \in SU(2) \quad (\text{unit quaternions}),$$

the *angular part*, multiplicative up to the ordering issues that Paper XXVII’s construction is built to handle: fixing the norm-ordering of the primes and, within a level, the self-conjugate generator sets of Paper XXVIII, the assignment $a \mapsto \rho(a)$ defines an ordered cocycle c_ρ on the Hurwitz tail

relation with values in the projective unit group $SU(2)/\{\pm 1\} \cong SO(3)$, and lifts to $SU(2)$ level by level.

Proposition 3.1 (The angular Kakutani series, exactly). *For the level- p generator set R_p , of cardinality $24(p+1)$ and closed under $a \mapsto -a$ and $a \mapsto \bar{a}$, the Sato–Tate average is exact: $\sum_{a \in R_p} \text{tr } \rho(a) = \frac{2}{\sqrt{p}} \sum_{a \in R_p} x_0(a) = 0$, by the symmetry $a \mapsto -a$. Hence*

$$\Xi_\beta(\rho) = \sum_p p^{-\beta} \sum_{a \in R_p} \|I - \rho(a)\|_{\text{HS}}^2 = \sum_p p^{-\beta} \cdot 4 |R_p| = 96 \sum_p (p+1) p^{-\beta},$$

with abscissa of convergence exactly 2 — the radial critical exponent $\beta_c(\zeta_P) = 2$ of Paper X. The angle turns critical precisely where the radius does. (Projectively, replacing ρ by its $SO(3)$ -image replaces the constant 4 by the corresponding adjoint-trace constant; the abscissa is unchanged.)

Proof. $\|I - u\|_{\text{HS}}^2 = 4 - 2 \text{Re tr } u$ on $SU(2)$; sum over R_p and use $-a \in R_p$, which cancels the trace term identically; $|R_p| = 24(p+1)$ is the Hurwitz–Jacobi count. Convergence of $\sum (p+1) p^{-\beta}$ is $\beta > 2$. $\square$

Remark 3.2 (The programme, with its hard step named). In the Gibbs phase $\beta > 2$ of Paper X, $\Xi_\beta(\rho) < \infty$; if Paper XXVII’s matrix-martingale coboundary theorem (its Theorem 4.3) transports from the independent geometric relation to the Hurwitz tail relation — the single hard step: the coordinates of the Hurwitz boundary are neither independent nor geometric, and the martingale filtration must follow the tree levels of Paper XXVIII — then c_ρ is a coboundary there and the KMS_β simplex of the angular system is $\text{Prob}(SU(2))$: a simplex over a genuinely nonabelian compact group, carried by an honest arithmetic semigroup, in its entire low-temperature phase. On the other side, the Lubotzky–Phillips–Sarnak spectral gap [3] makes every nontrivial representation of the level- p generators uniformly contractive, which is exactly the mixing hypothesis of Paper XXVII’s Theorem 4.5; where a scaling measure exists at the critical edge, the angular symmetry should therefore be fully restored. Verdict: the bottleneck theorem closes the scaling route universally; the angular route is open, constructive, and reduced to one named transport problem.

REFERENCES

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